

Exercise for Sect. 2.8

Exercise

2.8.1

Is the hypothesis that the sequence (a_n) be bounded necessary in Lemma 2.41.

Proof:

(2) $\Rightarrow$ (1) **fails** Let $J = \{k^2 : k \in \mathbb{N}\}$ is density zero, and $a_n = n$ for $n \in J$ otherwise 0.

The sequence converges to 0 for $n \notin J$ but the subsequence

$$A_{n^2+1} = \frac{1}{n^2+1} \sum_{j=0}^{n^2} a_j = \frac{1}{n^2+1} \frac{n(n+1)(2n+1)}{6} \rightarrow \infty.$$

Therefore, $\lim_{n \rightarrow \infty} \frac{1}{n} \sum a_j \neq 0$.

(2) $\Rightarrow$ (3) **fails** Same as above.

(1) $\Rightarrow$ (3) **fails** J is same as above but $a_n = n^{\frac{1}{4}}$ for $n \in J$. Then the first $m^2 + 1$

summation is $A_{m^2+1} = \frac{1}{m^2} \sum_{j=0}^{m^2+1} a_j = \frac{1}{m^2} \sum_{k=1}^m k^{\frac{1}{2}}$. The subsequence converges by integral test and it should converges to zero. Since the whole sequence is decreasing except $n \in J$, the sequence converges to zero.

But for squared sequence, $A_{m^2+1} = \frac{1}{m^2} \sum_{k=1}^m k = \frac{1}{m^2} \cdot \frac{m(m+1)}{2} \rightarrow \frac{1}{2}$. Therefore the whole sequence may not converge to zero.

(3) $\Rightarrow$ (1) **still true** By Cauchy-Schwarz inequality $\left(\frac{1}{n} \sum_{j=0}^{n-1} a_j\right)^2 \leq \frac{1}{n} \sum_{j=0}^{n-1} a_j^2$.

(1) $\Rightarrow$ (2) **still true** By the proof in textbook, it doesn't use the bounded condition.

(3) $\Rightarrow$ (1) **still true** Combine the two above.

□

Give an alternative proof of (1) $\implies$ (5) by proving the following statements:

1. Any factor of a weak-mixing transformation is weak-mixing.
2. A complex-valued eigenfunction f of U_T has constant modulus.
3. If f is an eigenfunction of U_T , then $x \mapsto \arg\left(\frac{f(x)}{|f(x)|}\right)$ is a factor map from $(X, \mathcal{B}, \mu, T)$ to $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, R_{\alpha})$ for some α .

Proof:

1. Suppose $(Y, \mathcal{B}_Y, \nu, S)$ is a factor of $(X, \mathcal{B}_X, \mu, T)$ with the factor map φ . Suppose $A, B \in \mathcal{B}_Y$, $A' = \varphi^{-1}(A)$, $B' = \varphi^{-1}(B)$ are in $\mathcal{B}_X$. By the assumption, $\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A' \cap T^{-n}B') - \mu(A')\mu(B')| \rightarrow 0$. Then,

$$\frac{1}{N} \sum_{n=0}^{N-1} |\nu(A \cap S^{-n}B) - \nu(A)\nu(B)| = \frac{1}{N} \sum_{n=0}^{N-1} |\mu(\varphi^{-1}(A \cap S^{-n}B)) - \mu(\varphi^{-1}(A))\mu(\varphi^{-1}(B))| \rightarrow 0$$

by the measure-preserving property of φ .

2. Suppose $U_T f = \lambda f$ almost everywhere. Take norm on the both sides and U_T is isometry, $\|f\|_2 = |\lambda| \|f\|_2$ implies $|\lambda| = 1$. Take absolute value pointwisely on the both sides yields $U_T |f(x)| = |U_T f(x)| = |\lambda| |f(x)| = |f(x)|$ almost everywhere. Thus, $|f|$ is T -invariant and it is constant by the ergodicity of T (from weak-mixing.)

3. By the arguments above, let $\lambda = e^{2\pi i \alpha}$ for some $\alpha \in \mathbb{T}$. If $|f| = 0$ almost everywhere, the map is trivial. Suppose $|f| = c > 0$, $\frac{f(x)}{|f(x)|} \in \mathbb{S}^1$ and let $e^{2\pi i \varphi(x)} = \frac{f(x)}{|f(x)|}$. Now prove that φ is the factor map. First, since $\varphi(x) = \arg\left(\frac{f(x)}{|f(x)|}\right)$, φ is automatically measurable. Second, since $U_T \frac{f}{|f|} = \lambda \frac{f}{|f|}$, $e^{2\pi i \varphi(Tx)} = e^{2\pi i \alpha} e^{2\pi i \varphi(x)} = e^{2\pi i (\varphi(x) + \alpha)}$. This proves $\varphi \circ T = R_{\alpha} \circ \varphi$. Since Last, let $\nu = \mu \circ \varphi^{-1}$ be the push-forward measure. For a measurable set $A \in \mathcal{B}_{\mathbb{T}}$ and $t \in \mathbb{T}$, $\nu(t + A) = \mu(\varphi^{-1}(t + A)) = \mu()$

□

Exercise

2.8.3

If a measure-preserving system $(Y, \mathcal{B}_Y, \nu, S)$ is not totally ergodic then there exists a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and $K > 1$ with the property that $(Y, \mathcal{B}_Y, \nu, S)$ is measurably isomorphic to the system

$$(X^{(K)}, \mathcal{B}^{(K)}, \mu^{(K)}, T^{(K)}).$$

Proof: First suppose S is ergodic. Suppose K is the smallest integer such that S^K is not ergodic. Since S^K is not ergodic, $U_{S^K} = (U_S)^K$ contains non-constant eigenfunction g such that $(U_S)^K g = g$. Let $\lambda = e^{\frac{2\pi i}{K}}$ and $U_S g = \lambda g$. Since $U_S |g| = |U_S g| = |\lambda| |g| = |g|$, $|g|$ is S -invariant and then is constant almost everywhere. Without loss of generality, assume $|g| = 1$ a.e.

For g^K , $g^K(Sy) = (g(Sy))^K = (\lambda g(y))^K = g^K(y)$ is S -invariant. Thus, $g^K = 1$ a.e. and $g(y) \in \{\lambda^k : k \in [0, K-1] \cap \mathbb{N}\}$ a.e. Let $A_k = g^{-1}(\{\lambda^k\})$ for $k \in [0, K-1] \cap \mathbb{N}$ and then $\{A_k\}$ is a partition of Y . The sets also have the cyclic property, which is, $S^{-1}A_k = A_{k+1 \bmod K}$. By S is measure-preserving, $\nu(A_k) = \frac{1}{K}$ for all k .

Let $X = A_0$, $\mathcal{B} = \mathcal{B}_Y|_{A_0}$, $\mu = K\nu|_{A_0}$, and $T = S^K$. And ...

□

Exercise**2.8.4**

For a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and function $f \in L^2_\mu$, show that the function $n \mapsto \langle U_T^n f, f \rangle$ is positive-definite. Apply the Herglotz-Bochner theorem to translate the problem into one concerned with function on $\mathbb{S}^1$, and there use the fact that $\frac{1}{N} \sum_{n=1}^N \rho^n$ converges for $\rho \in \mathbb{S}^1$.

Proof: For any $c_1, c_2, \dots, c_k \in \mathbb{C}$ and $n_1, n_2, \dots, n_k \in \mathbb{N}$,

$$\begin{aligned}
 \sum_{i=1}^k \sum_{j=1}^k c_i \overline{c_j} \langle U_T^{n_i - n_j} f, f \rangle &= \sum_{i=1}^k \sum_{j=1}^k \langle c_i U_T^{n_i} f, c_j U_T^{n_j} f \rangle \\
 &= \left\langle \sum_{i=1}^k c_i U_T^{n_i} f, \sum_{j=1}^k c_j U_T^{n_j} f \right\rangle \\
 &= \left\| \sum_{i=1}^k c_i U_T^{n_i} f \right\|_2^2 \geq 0.
 \end{aligned}$$

□